

DTSA 5304

Fundamentals of Data Visualization

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 2 – Vital Skills for Data Scientists	Machine Learning Engineer (79.4 / 100)	56.2 / 100
SPECIALIZATION Vital Skills for Data Scientists		

56

CURRENT AVG SCORE
across 6 job roles

91

POTENTIAL SCORE
with recommended additions

+55

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Explores the design, development, and evaluation of information visualizations. Combine aspects of design, computer graphics, HCI, and data science, to gain hands-on experience with creating visualizations, using exploratory tools, and architecting data narratives. Topics include user-centered design, web-based visualization, data cognition and perception, and design evaluation.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	79.4	100.0	+20.6	MEDIUM
AI Engineer	60.5	88.1	+27.7	MEDIUM
ML Engineer	59.2	86.4	+27.2	MEDIUM
Data Scientist	51.7	92.8	+41.1	HIGH
MLOps Engineer	47.4	87.8	+40.4	HIGH
Applied Scientist	39.1	93.7	+54.6	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

LLMs & Generative AI 2 of 6 roles

storage

Experiment Tracking 1 of 6 roles

tracking

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- HIGH** **Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- MEDIUM** **Consider LLMs & Generative AI content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: storage.
- LOW** **Optional: Experiment Tracking** — would broaden appeal to one additional role. Candidate terms: tracking.
- LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

79.4 → 100.0 (+20.6 pts)

predictive web learning analytics business models model applications

AI Engineer

60.5 → 88.1 (+27.7 pts)

models learning technical model systems problems storage machine

ML Engineer

59.2 → 86.4 (+27.2 pts)

learning machine systems models model world problems new

Data Scientist

51.7 → 92.8 (+41.1 pts)

analytics analytical problems

MLOps Engineer

47.4 → 87.8 (+40.4 pts)

model systems learning machine tracking models

Applied Scientist

39.1 → 93.7 (+54.6 pts)

lab computational research storage interdisciplinary problems models technical

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.